Google Cloud H2S

AI Builder Cup

Build in JAPAC.
Build for the world.



## Team Details

Scam Shield

a. Team name: **Nexustrip**

b. Team leader name: **Naman Shrimal**

c. Problem Statement:
**BFSI: Intelligent Risk, Fraud & Financial Experiences**
People in India lose crores to scam messages, QR codes and calls that look real. Scam Shield lets anyone check what they were sent in seconds, then act: a complaint draft, a shared scam list that warns the next person, and a WhatsApp card for the family. Community benefit: fewer people lose their savings, and every report protects the next one.

# Brief about the idea

## Check what you were just sent before you act on it.

Free for anyone in India, and built for the scams people meet here: fake payment receipts, "digital arrest" calls and UPI QR tricks.

### One multimodal Gemini call

Text, screenshots, QR codes, links and Hinglish voice notes are read together by Gemini on Vertex AI.

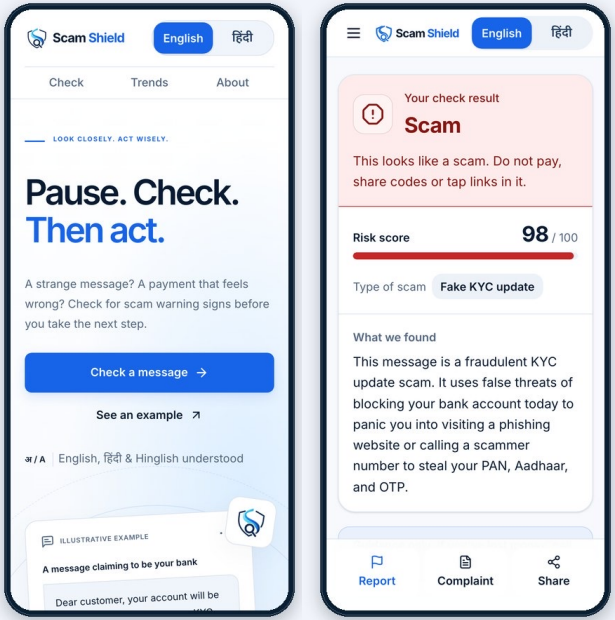
### A verdict you can check

Scam, Suspicious or No scam signs found. Every red flag quotes the exact words. English or Hindi.

### From verdict to action

A complaint draft for 1930 and cybercrime.gov.in, a shared scam list and a WhatsApp warning card.

**Careful by design:** links open in an isolated browser, never on your phone. It never says "safe": it gives guidance and you decide.




The live app, as seen on a phone.

# Opportunities

**Rs 19,813 crore** reported lost to cyber fraud in 2025, across about 21.78 lakh complaints (I4C data, National Cyber Crime Reporting Portal)

## How different is it from existing ideas?

**Truecaller Scam Checker (Sep 2026):** checks links, numbers and messages against community reports; screenshot upload is only planned.

**Google scam detection:** Messages on Pixel 6 and newer, and calls on Pixel 9 and newer, in India.

**Chakshu (Sanchar Saathi):** a reporting facility for suspicious calls, SMS and WhatsApp messages.

**Scam Shield:** reads the content itself, so it can flag scams nobody has reported yet, from text, screenshots, QR codes, links and voice notes.

## How will it solve the problem?

- Checks before you pay, scan or click
- Quotes every red flag from the message
- Turns a scam into action: complaint, shared list, family warning

## USP of the proposed solution

- Catches scams nobody has reported yet
- Made for India: fake payment receipts, "digital arrest" calls, UPI QR tricks
- Understands Hinglish voice notes

# List of features offered by the solution

### 5 input types
Text, screenshot, QR code, link and voice note.

### Clear verdict
Scam, Suspicious or No scam signs found, with a risk score.

### Evidence quoted
Each red flag points to the exact words in the message.

### English + Hindi
Hinglish understood; answers in the language you pick.

### Complaint draft
Ready text for 1930 and cybercrime.gov.in.

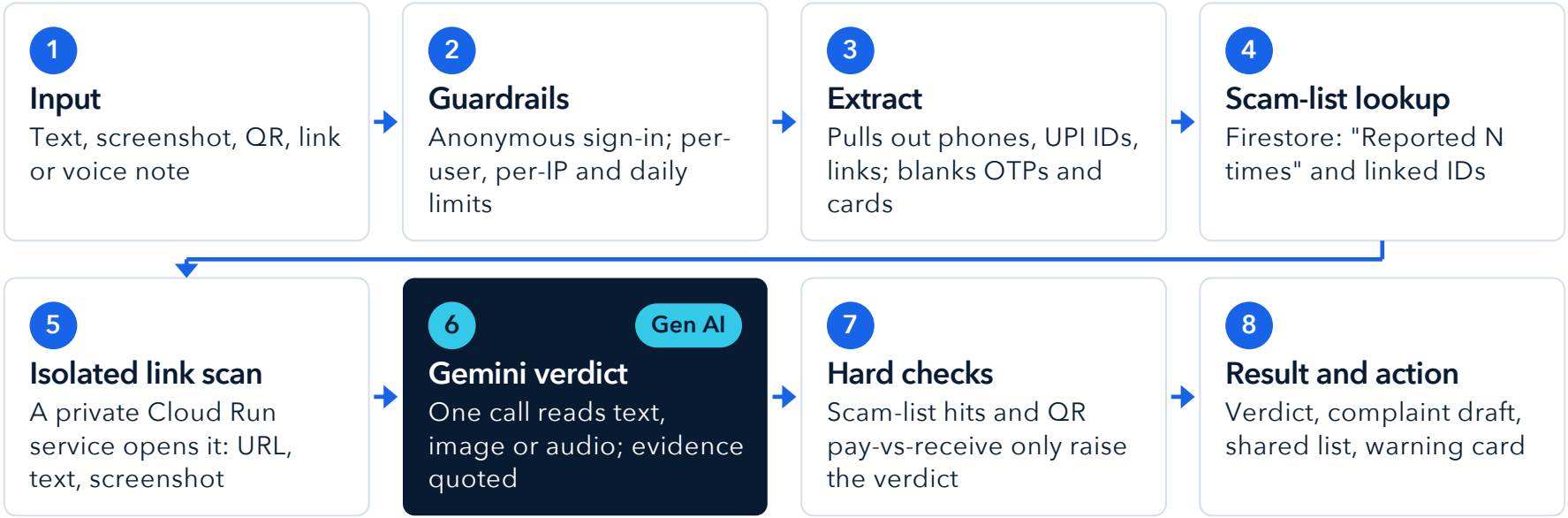
### Shared scam list
"Reported N times", with linked numbers, UPI IDs and sites.

### Warning card
One tap shares a warning with the family WhatsApp group.

### Agent tools
WebMCP tools let AI agents check and report.

**Also:** trends, history, one-tap examples, an installable app and the Android share menu.

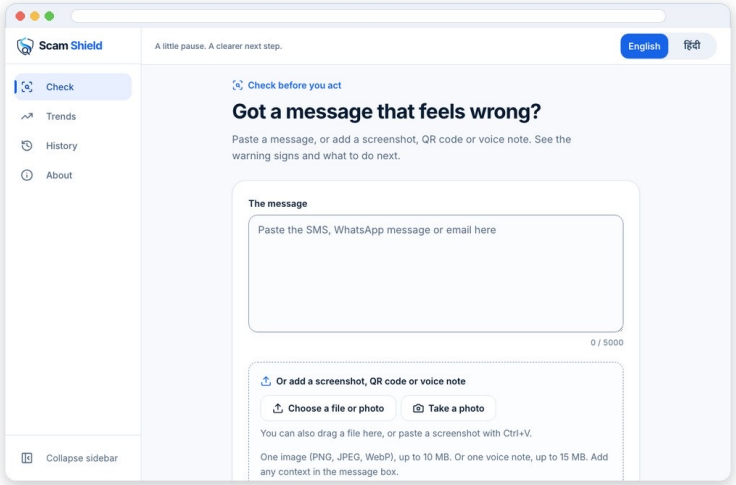
# Process flow diagram or Use-case diagram



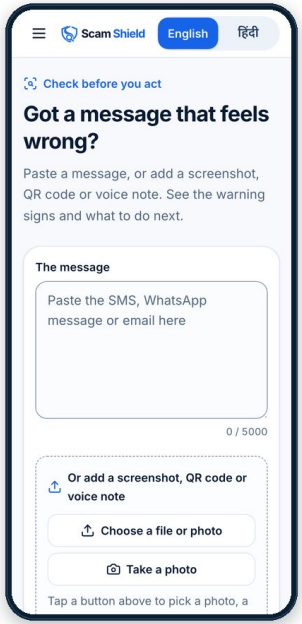

**Why Gen AI:** rules cannot read a screenshot or a Hinglish voice note, or judge a scam nobody has reported. Gemini does. The hard checks can only raise its verdict, never lower it.

If Gemini fails or times out, the person sees a clear retry message; no verdict is invented and nothing is stored.

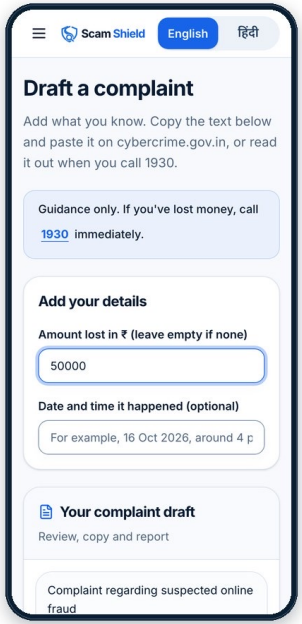
# Wireframes/Mock diagrams of the proposed solution (optional)




**Web app**
The same flow on desktop, with a sidebar.




**1 Paste or share**
Text, photo, QR or voice note.

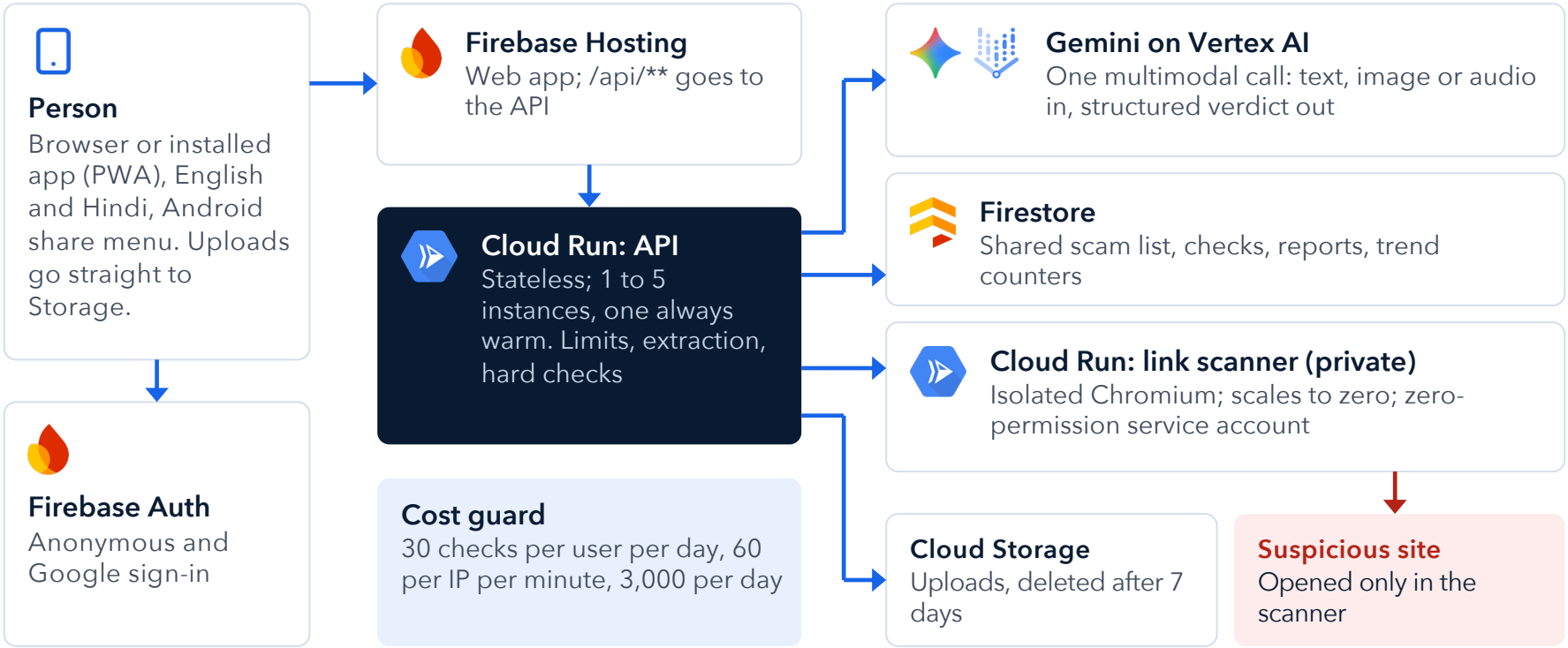



**2 Act on a scam**
A complaint draft ready to file.




**3 Hindi**
The whole app switches language.

# Architecture diagram of the proposed solution




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# Technologies to be used in the solution

## Gen AI that does real work

- **Gemini 3.7 Flash** on Vertex AI, with Gemini 2.5 Flash as automatic fallback
- **One multimodal call** reads text, screenshots and audio (Hinglish voice)
- **Schema-validated output;** every red flag quotes evidence; English or Hindi
- **Hard checks only raise** the verdict: scam list, QR pay-vs-receive
- **Does what rules cannot:** unreported scams, images, voice
- **WebMCP agent tools:** check_message and report_scam

## Google Cloud and Firebase, built to scale

- **Cloud Run API:** stateless and autoscaling
- **Link scanner:** scales to zero, own zero-permission service account
- **Firestore** shared scam list; **Firebase** Hosting, Auth, Storage

## Application

React 18 | Vite | Tailwind | Workbox PWA | Node 22 | TypeScript | Hono | zod | Playwright | sharp + jsQR

**590+ automated tests**, plus a 185-case live benchmark

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## Estimated implementation cost (optional)

FIXED

### about US$13

per month (estimate)

One Cloud Run API instance kept warm (1 vCPU, 1 GiB, Mumbai) at US$0.0000025 per vCPU-second and per GiB-second, before any free tier.

PER CHECK

### US$0.004–0.008

per text check (estimate)

Gemini 3.7 Flash at about 2,500 input and 500 to 1,500 output tokens. US$0.75 in, US$3.75 out per 1M tokens to 31 Dec 2026; double after.

FREE TIERS

### Free tiers

expected at this volume (estimate)

Firebase Hosting, Auth, Firestore and Storage are expected to stay inside free tiers. The link scanner bills only while it scans.

**Limits are the real control**

30 checks per user per day, 60 per IP per minute, 3,000 per day overall. These cap spending.

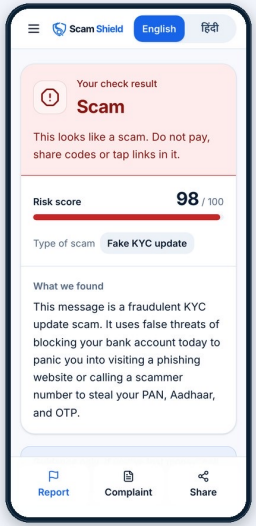
**Ceiling at the daily cap**

3,000 checks at the per-check estimate is about US$11 to 23 per day for Gemini.

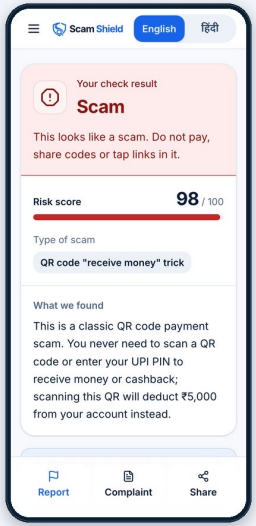
**Budget alert: Rs 2,000 a month**

Alerts at 50%, 90% and 100% (Rs 1,000, Rs 1,800, Rs 2,000). An alert notifies; it does not stop spending.

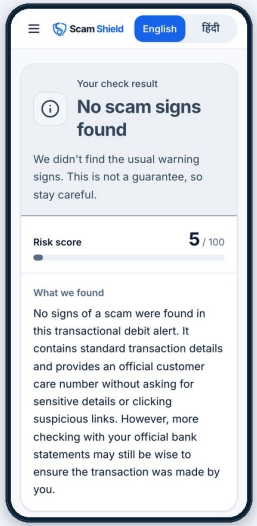
# Snapshots of the prototype



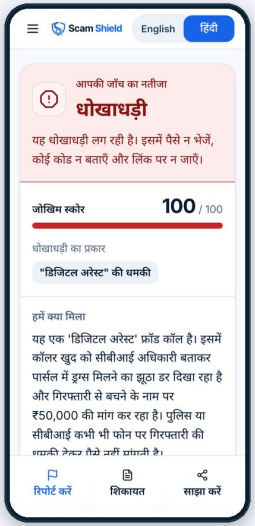

**Scam**
Fake KYC SMS, evidence quoted



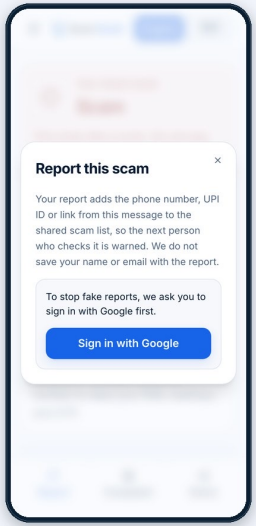

**Scam**
QR asks you to pay, text says receive




**No scam signs found**
A genuine bank alert; not a guarantee




**Scam, in Hindi**
Hinglish voice note, fake police call




**Report**
Adds it to the shared scam list

**Live benchmark,** 185 cases over 3 runs: 98.3% accuracy, 100% scam recall, 3.3% false positives (3 of 90 genuine)

# Prototype Performance report/Benchmarking

| 98.3% | 100% | 3.3% | 96.7% |
|---|---|---|---|
| Accuracy | Scam recall | False-positive rate | Scam type correct |

185 synthetic cases over 3 live runs: 90 scams, 90 genuine, 5 ambiguous (not scored). Accuracy is 177 of 180 scored.

| Run | Cases | Accuracy | Scam recall | False positives |
|---|---|---|---|---|
| **v1 original** | 50 | 100% | 100% | 0% |
| **v2 harder cases** | 35 | 93.3% | 100% | 12.5% |
| **v3 realistic** | 100 | 99.0% | 100% | 2.0% |
| **Combined** | **185** | **98.3%** | **100%** | **3.3%** |

**Latency:** text p50 4.5 s, p95 18.2 s; image p50 6.9 s; voice 10.3 s. **Errors:** 0. p50 is the median, p95 the slowest 5%.

**Keyword baseline:** on v2's 34 text cases it scored 94.1%, the same as Gemini, with different mistakes; v3 84.0% accuracy and 72.0% recall (Gemini 99.0% and 100%).

**First run,** 50 cases: 100% accuracy, 100% recall, 0% false positives. **Hindi run,** same 50 cases: 100%, 100%, 0%.

## Read this honestly

- **185 cases, written by the authors:** a test, not a guarantee. 90 scams over 13 types, 90 genuine.
- **3 false positives:** scam-awareness forward that quotes a scam link, delivery update with a tracking link, friend asking for a short loan. All misses and false positives: eval/results/combined.md.
- **Why gemini-3.7-flash:** an earlier 50-case run on gemini-3.8-flash scored 98.0% accuracy with 4.2% false positives, image checks up to 34 s.

# Additional Details/Future Development (if any)

### More Indian languages
Tamil, Telugu, Marathi, Bengali and more.

### Bank & telecom partners
Verified reports feed the shared scam list.

### URL reputation feeds
Add Google Web Risk next to the isolated scan.

### Android app
Share SMS and notifications to a check.

### Google Search grounding
Switch on for link checks once quota allows.

### Call-recording upload
Check recorded calls, not just voice notes.

### Misinformation mode
The same engine for "is this forward real?"

**Plans, not shipped.**
Search grounding is built but off by default.

## How it scales

**Stateless API**
Cloud Run adds instances (1 to 5 today) as traffic grows; the daily cap is a setting, not a rebuild.

**Managed data**
Firestore, Hosting, Auth and Storage need no servers to run; uploads auto-delete after 7 days.

**Cost stays bounded**
Per-user, per-IP and global limits rise only as budget allows; App Check adds abuse protection next.

## Provide links to your:

- **GitHub Public Repository**
  https://github.com/Namans12/scam-shield
- **Demo Video Link (3 Minutes)**
  https://youtu.be/a678nAYddq0
- **Final Product Link**
  https://scam-shield-live.web.app

**Scan to try the live app**

Paste a suspicious message and see the verdict.



# Scam Shield

## Pause. Check. Then act.

Check what you were just sent before you act on it. Free, in English and Hindi.

- Gemini reads any input
- Evidence for every flag
- A scam becomes action

scam-shield-live.web.app

Google Cloud H2S

# AI Builder Cup

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# Thank you

